

NAVEEN KUMAR UPPALAPATI, PhD

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PROFESSIONAL SUMMARY

Research Scientist with 8+ years advancing embodied intelligence and robotic manipulation in unstructured environments. Expertise in dexterous manipulation, multimodal sensor fusion (vision, tactile, proprioception), and vision-based control systems. Strong publication record with first-author papers at RSS, ICRA, and IEEE RA-L. Proven ability to design, build, and deploy complex robotic systems from concept to real-world operation. Seeking to contribute to cutting-edge research in robotics foundation models and embodied AI.

EDUCATION

PhD, Systems and Entrepreneurial Engineering

University of Illinois Urbana-Champaign (UIUC)

May 2020

Urbana, IL, USA

- Dissertation: Design and Characterization of Soft Continuum Manipulators using Fiber Reinforced Actuators

Focus: Dexterous manipulation, control, Reinforcement Learning (RL)

MS, Systems and Entrepreneurial Engineering

University of Illinois Urbana-Champaign (UIUC)

December 2016

Urbana, IL, USA

B.Tech, Instrumentation and Control Engineering

National Institute of Technology (NIT), Tiruchirappalli, India

August 2013

- First Rank Award (2011, 2012, 2013) — Institute Medal — IOCL Scholarship

CORE TECHNICAL EXPERTISE

Robotics & Embodied Intelligence: Dexterous Manipulation, Multimodal Sensor Fusion (Vision, Proprioception), Vision-Based Control, Compliant Actuation, Real-World Deployment

Programming & Software: Python (Expert), C++, MATLAB, ROS/ROS2, PyTorch, TensorFlow

Computer Vision & Perception: Visual Servoing, Multi-Camera Systems (RGB, Depth), 3D Perception, SLAM

Machine Learning: Deep RL for Control, Computer Vision Applications, Vision-Language Models (Emerging)

Simulation & Tools: Gazebo, MuJoCo

PROFESSIONAL EXPERIENCE

Research Scientist

Center for Digital Agriculture (CDA), NCSA, UIUC

June 2022 – Present

Urbana, IL

- Developed dual-camera perception system for precision manipulation achieving 92% detection accuracy in occluded scenes (IEEE ICRA 2025)
- Built end-to-end robotic systems combining perception, planning, and control for autonomous operation in real-world unstructured environments
- Secured \$9M+ in competitive research funding (USDA); led cross-functional teams (hardware, software, ML) developing autonomous manipulation systems from conception through deployment
- Conducted large-scale field testing (40+ hours) of robotic systems, systematically analyzing failure modes and iterating toward robust real-world performance
- Fostered collaborations with industry partners and academic researchers across multiple institutions

Research Engineer

Agricultural and Biological Engineering, UIUC

March 2022 – June 2022

Urbana, IL

- Developed vision-based methods for real-time control and 3D shape reconstruction of deformable robotic manipulators, enabling precise pose control in dynamic environments
- Mentored graduate and undergraduate researchers in robot design, sensor integration, and control system development

Postdoctoral Research Associate

Industrial & Systems Engineering + Coordinated Science Lab, UIUC

March 2020 – March 2022

Urbana, IL

- Designed, built, and demonstrated semi-autonomous manipulation robot combining hybrid soft-rigid arm with vision-based control for delicate object handling in cluttered environment (RSS 2020)
- Achieved <3cm positioning accuracy through visual servoing and adaptive control, demonstrating 78% success rate across 60 real-world manipulation trials

Graduate Research Assistant

Industrial & Systems Engineering, UIUC

January 2014 – December 2019

Urbana, IL

- Developed deep reinforcement learning-based control algorithms for soft continuum manipulators, achieving <3cm accuracy in reaching tasks through model-free learning
- Created novel morphological switching concept for hybrid soft-rigid arms, demonstrating 5× increase in payload capacity while maintaining compliance for safe manipulation
- Designed parallel asymmetric soft continuum arms and grippers for dexterous manipulation of delicate objects

KEY PUBLICATIONS (Selected from 18+ Total)

- **K. Kendall et al., N.K. Uppalapati, G. Chowdhary, G. Krishnan**, “Precision harvesting in cluttered environments: Integrating end effector design with dual camera perception,” *IEEE ICRA*, 2025
- **J. Cuaran et al., N.K. Uppalapati, G. Chowdhary**, “Under-Canopy Dataset for Advancing Simultaneous Localization and Mapping in Agricultural Robotics,” *International Journal of Robotics Research*, 2023
- **S.K. Kamtikar et al., N.K. Uppalapati, G. Krishnan, G. Chowdhary**, “Visual Servoing for Pose Control of Soft Continuum Arm in a Structured Environment,” *IEEE Robotics and Automation Letters*, March 2022
- **S.H. Kim et al., N.K. Uppalapati**, “A physics-informed, vision-based method to reconstruct all deformation modes in slender bodies,” *IEEE ICRA*, 2022
- **N.K. Uppalapati et al., G. Chowdhary, G. Krishnan**, “A Berry Picking Robot with a Hybrid Soft-Rigid Arm: Design and Task Space Control,” *Robotics Science and Systems (RSS)*, 2020
- **K. Koe et al., N.K. Uppalapati, G. Krishnan, G. Chowdhary**, “Learning based Position and Orientation Control of a Hybrid Rigid-Soft Arm Manipulator,” *ASME Journal of Mechanisms and Robotics*, Feb 2025

Complete record: 7 Journal papers, 11 Conference papers (ICRA, RSS, RoboSoft); 50+ peer reviews

AWARDS & RECOGNITION

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| • 3rd Place, Soft Robotics Competition, RoboSoft 2018, Italy | • ISE Award (\$1,888), UIUC, 2019 |
| • Best Robot Demo, CSL Student Conference, UIUC, 2018 | • Institute Medal, NIT Trichy, 2013 |
| • 3rd Place, ASME Robot Design Competition, 2015 | • First Rank Award, NIT Trichy (2011, 2012, 2013) |

RESEARCH IMPACT & LEADERSHIP

Funding & Program Leadership: Secured and managed \$9M+ in competitive research funding (USDA). Led interdisciplinary teams across hardware, software, and machine learning domains. Established collaborations with industry partners and research institutions across multiple projects.

Real-World System Deployment: 40+ hours autonomous operation in unstructured, real-world environments. Systematic failure analysis and iterative improvement methodology. Achieved 78% success rate in manipulation tasks under challenging conditions (occlusions, variable lighting, cluttered scenes).

Technical Mentorship: Mentored 15 graduate students and 13 undergraduate researchers in robotics system design and development. Supervised students who progressed to top PhD programs and industry positions. Collaborative research resulting in 12 co-authored publications.

Publication Impact: 18+ peer-reviewed publications at premier venues (RSS, ICRA, IEEE RA-L, IJRR). First-author paper at Robotics: Science and Systems (RSS 2020) on dexterous manipulation. 50+ peer reviews for top robotics conferences and journals.

PROFESSIONAL SERVICE & OUTREACH

Workshop Organizer: Agricultural Robotics for a Sustainable Future (IROS 2023, 2024, 2025)

Editorial Service: Review Editor, Frontiers in Robotics and AI - Soft Robotics (Since 2023)

Peer Review: 50+ reviews for ICRA, RSS, IROS, IEEE RA-L, and other premier venues

Teaching: Guest Lecturer (Soft Robotics); TA for Control Systems, Operations Research (Rated Excellent)

References available upon request